

Credit Card Default Prediction

Risk, interpretation and decision costs

Emerging Topics in FinTech

THE TEAM

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Contribution: 25% each

How we split and use the data

30,000 clients • 23 predictors •
22.12% default

Categories learned within each
training fold

Validation selects policies; historical
test reports outcomes

Partition and purpose

Fitting	18,000
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Five-fold CV and final fitting

Validation	6,000
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Select decision thresholds

Test	6,000
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Evaluate frozen decisions

Source: data/splits; protocol_frozen.json; results/final/audit.json

Logistic regression: the linear reference

L2 model + fold-trained scaling + one-hot categories

12 declared settings; CV selected
C=0.1, unweighted

Original LR notebook was empty;
original process unverified

AP 0.4967

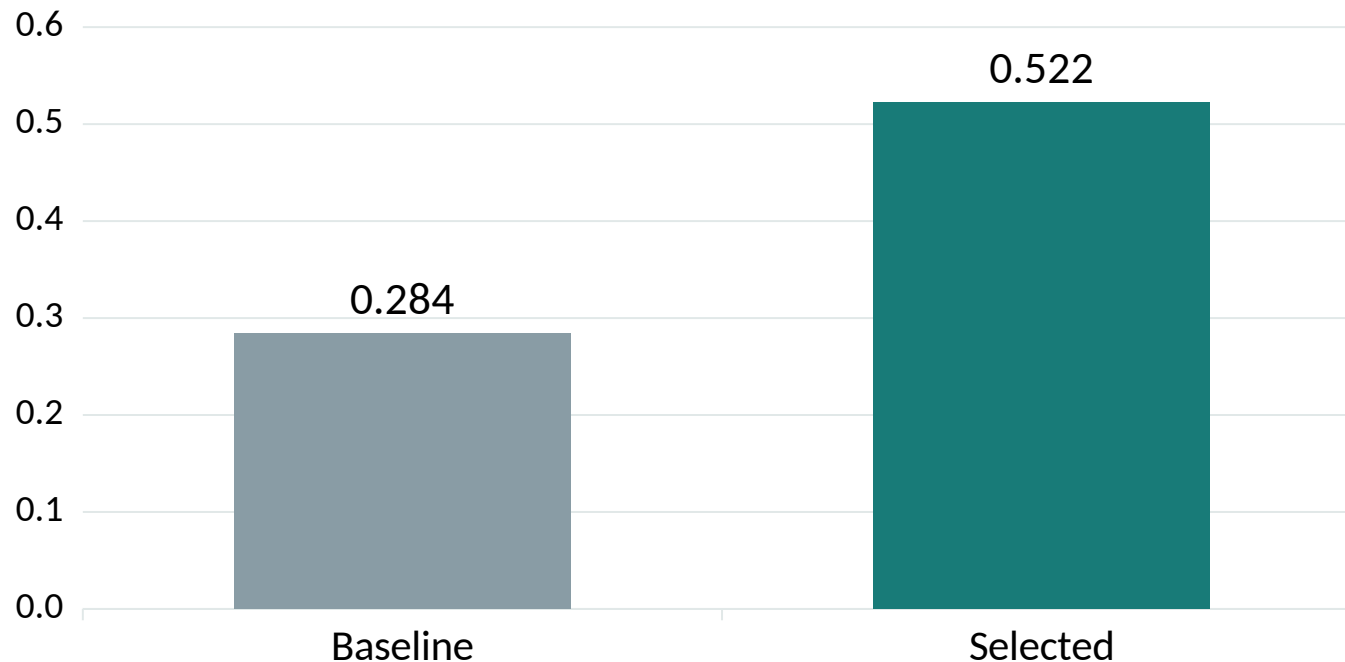
Baseline AP 0.4970

No ranking improvement from selection
LR* is a separate post-hoc supplement

Source: results/final/model_comparison.csv

Decision tree: controlling overfitting

Test average precision



21 configurations: depth × minimum leaf size

Unrestricted baseline fits training folds almost perfectly

Selected minimum leaf size: 100 observations

A large baseline gap signals overfitting.
CV selection improves this tree's ranking.

Source: results/final/model_comparison.csv

Reading the tree's first decision

Root rule: recent repayment status
 $\text{PAY_0} \leq 1.5$

Selected tree: depth 18, with 134
leaves

Node proportions describe this fitting
sample

Recent repayment status

$\text{PAY_0} \leq 1.5$

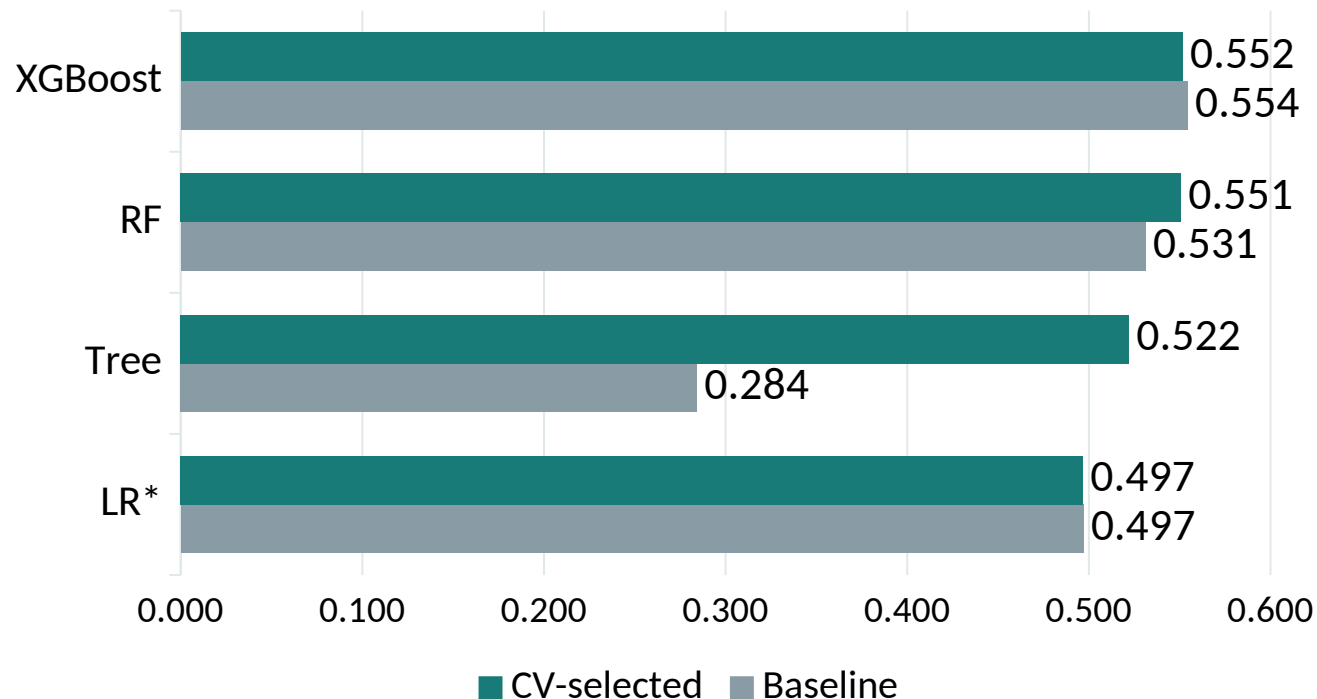
Yes · 16.51% No · 70.13%

Default proportion in each fitting-sample branch.
These are associations, not causal effects.

Source: saved decision-tree model; integration audit/report evidence.

Ranking performance across four models

Test average precision



AP emphasizes positive-class retrieval

AP is not trapezoidal PR-AUC

CV selection precedes these test comparisons

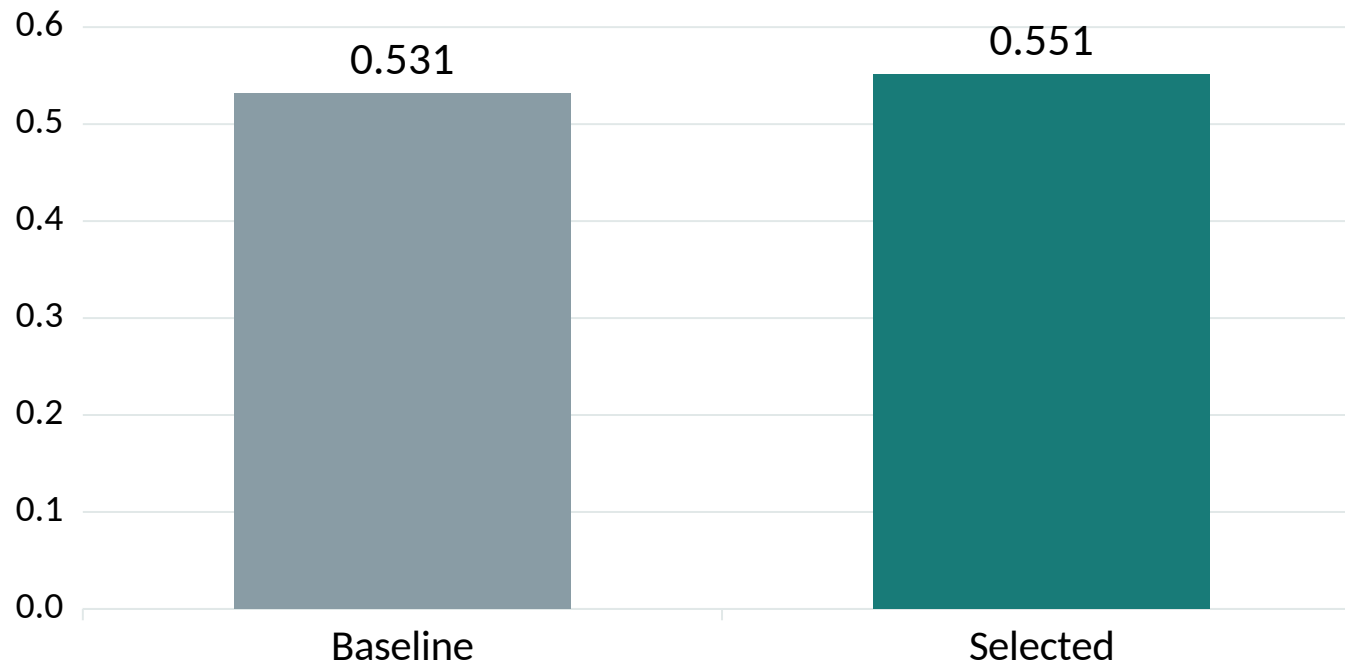
Baseline and selected models are both retained.

Small ensemble differences do not establish superiority.

Source: results/final/model_comparison.csv

Random forest: averaging and regularization

Test average precision



Baseline + 23 seeded candidate settings

Selected: 500 trees, depth 8, minimum leaf 2

Positive-class weight 3; feature fraction 0.5

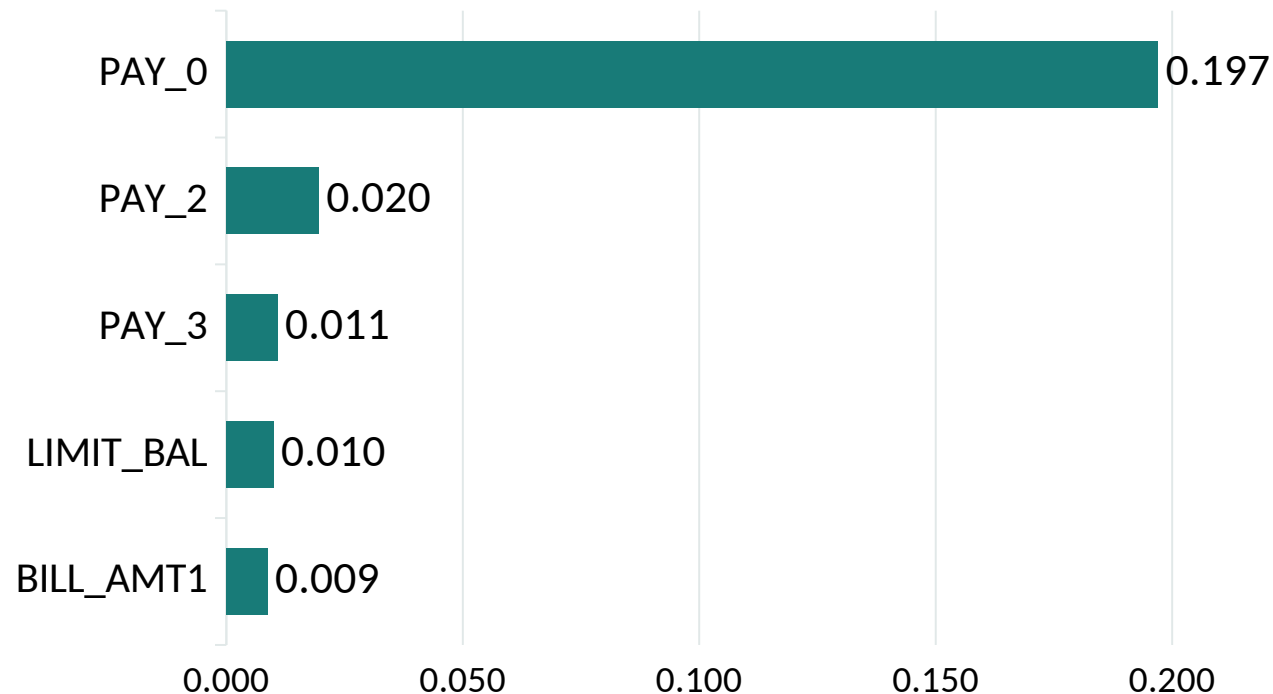
Averaging reduces dependence on one tree.

Weights also change the score distribution.

Source: results/final/model_comparison.csv

Random forest: which features matter?

Mean decrease in validation AP



PAY_0 has the largest validation AP decrease

Permutation measures predictive reliance

Correlated variables can substitute for each other

Five saved shuffles per feature.

Predictive association, not causality.

Source: results/rf/validation_importance.csv

Random forest: two illustrative errors

At 0.5: 637 missed defaults; 573 false alarms

Missed default · row 982

Score 0.1012; actual default = 1

False alarm · row 28747

Score 0.9415; actual default = 0

Source: results/rf/misclassification_cases_05.csv

Why the predictions can fail

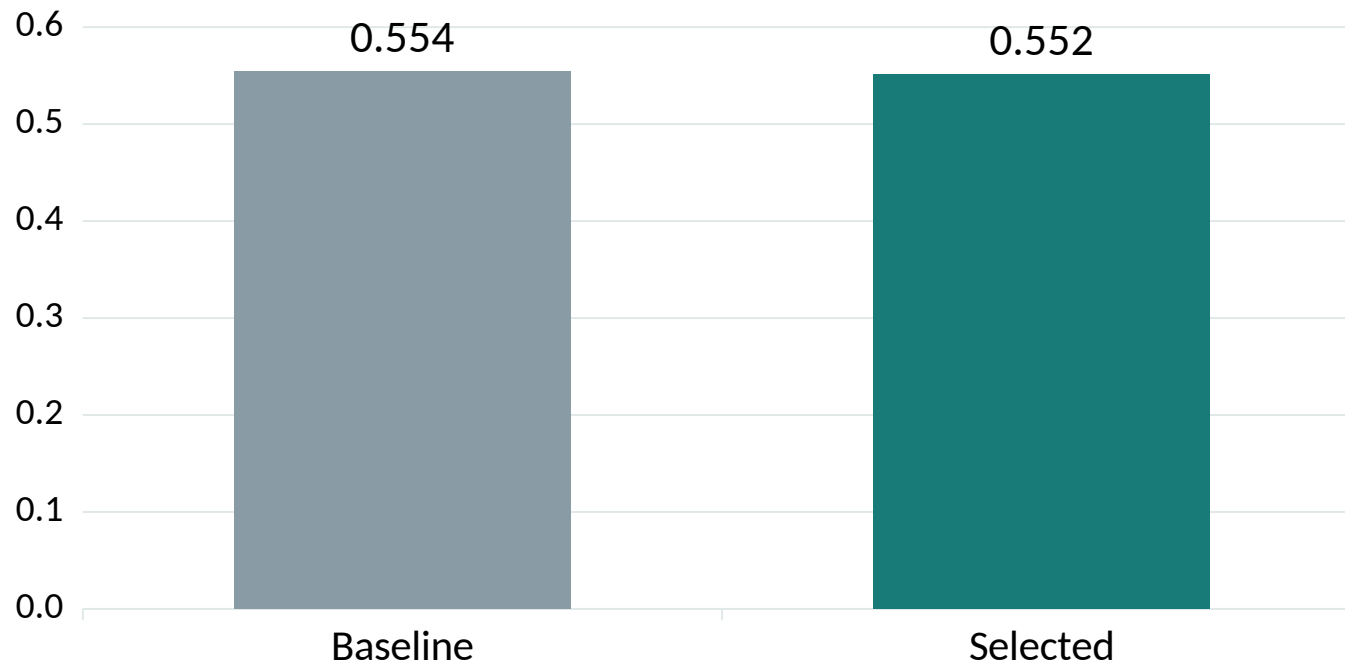
Missed default: PAY_0 = -2

False alarm: PAY_0 = 3

Illustrative extreme errors, not typical clients.

XGBoost: no gain in test ranking

Test average precision



24 configurations; select by CV AP

Selected: 150 trees, depth 4, rate 0.03

Selected ROC-AUC: 0.7762

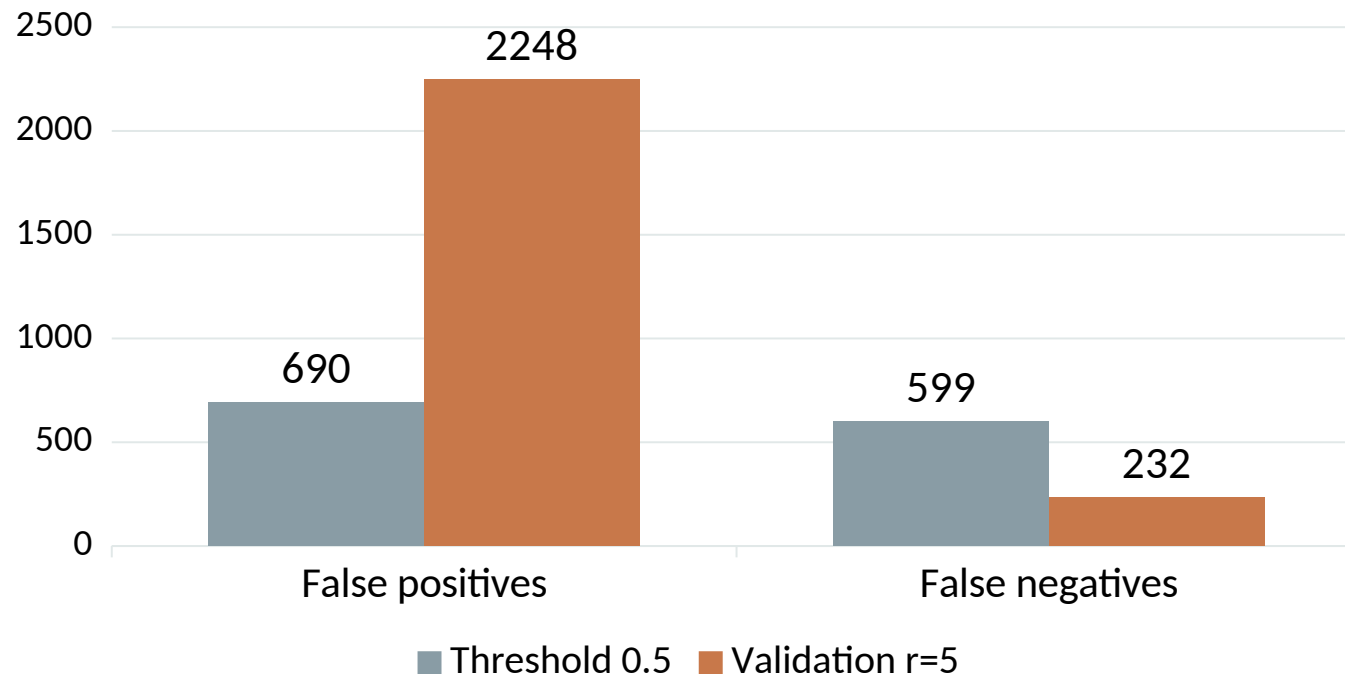
Tuned – baseline AP: -0.0028

Conditional 95% interval: [-0.0106, 0.0054]

Source: results/final/model_comparison.csv

XGBoost: fewer misses, more reviews

Test confusion trade-off



Validation r=5 cutoff: 0.3150

Recall 54.9% → 82.5%; alerts 55.7%

Cost 3,685 → 3,408: 7.5% reduction

Same ranking scores, different action policy.

Review costs are hypothetical; capacity is unmeasured.

Source: results/final/model_comparison.csv

What the results mean for credit review

Ranking, explanations, and workload need separate evidence

No clear ensemble winner from these small differences

Next: fresh temporal data, calibration, subgroup errors

Discussion

Set realistic costs and review capacity.
Validate the policy on fresh data.

Historical educational study; no causal or deployment-readiness claim.

Backup A • AP, ROC-AUC and threshold metrics

AP = sum of recall increments × precision

ROC-AUC measures positive/negative score ordering

Precision, recall, F1 and accuracy require a threshold

Two different definitions

Saved continuous scores supply ranking metrics.

Changing the cutoff changes decisions only.

Source: [scikit-learn average_precision_score documentation](#); part4/evaluation.py

Backup B • Cost ratios and fixed-prediction uncertainty

Predeclared ratios: $r = 1, 3, 5, 10$

Validation minimum cost; ties prefer largest cutoff

1,000 stratified paired bootstrap resamples

What the intervals cover

Fixed models / thresholds / class counts

Excludes fitting / selection uncertainty

Excludes population / prevalence shifts

Source: frozen protocols; integration/uncertainty.py; bootstrap_intervals.csv

Backup C • Data provenance and evidence boundaries

UCI cleaning reproduced exactly from downloaded source

EDUCATION 0/5/6 → 4; MARRIAGE
0 → 3

Retained duplicate vectors can span partitions

Sources: results/final/source_audit.json; integration/LOGISTIC_PROVENANCE.md

The limits of historical evidence

Original LR process remains unverified.

LR* source was committed before its one run.

No new unseen-test claim is made.

Backup D • Sources, responsibilities and reproducibility

UCI: doi.org/10.24432/C55S3H

scikit-learn: [average_precision_score](https://scikit-learn.org/stable/modules/average_precision_score.html)
documentation

XGBoost paper:
doi.org/10.1145/2939672.2939785

Member contributions • 25% each

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Part 1 — Data and logistic regression

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Part 2 — Decision tree and overfitting

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Part 3 — Random forest and error analysis

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Part 4 — XGBoost and threshold trade-offs

Sources: Group 5 repository, frozen protocols, row-level predictions and integration audit.